

Speech Recognition Based On Transformer Neural Networks

N.S. Mamatov

Research institute for development
of digital technologies and artificial
intelligence,

Tashkent, Republic of Uzbekistan.
m_narzullo@mail.ru

A.N.Samijonov

Tashkent University of Information
Technologies named after Muhammad
al-Khwarizmi,

Tashkent, Republic of Uzbekistan
an_samijonov@mail.ru

N.A.Niyozmatova

Tashkent University of Information
Technologies named after Muhammad
al-Khwarizmi,

Tashkent, Republic of Uzbekistan
n_nilufar@mail.ru

K.K.Erejepov

Tashkent University of Information
Technologies named after Muhammad
al-Khwarizmi,

Tashkent, Republic of Uzbekistan
e_keulimjay@mail.ru

Sh.Sh. Abdullaev

Research institute for development
of digital technologies and artificial
intelligence,

Tashkent, Republic of Uzbekistan
shash3005@gmail.com

Abstract—This article is devoted to the creation of a speech recognition system in the Uzbek language based on the analysis of existing approaches to speech recognition and the selection of the preferred approach, which provides information on the End2End approach and its effective use. In addition, several manifestations of transformer neural networks have been analyzed. Based on the results of the analysis, high quality and efficiency were used in speech recognition in Uzbek. The results obtained showed that transformer-based models were superior to other widely used models in terms of training time and accuracy.

Keywords—speech, signal, feature, transformer, attention, encoding, natural language processing, recognition

I. INTRODUCTION

Transforming speech to text is a multi-step process of acoustic signal analysis, which, while seem simple, is a complex process that requires a great deal of labor. According to an analysis by experts from Gartner, a UK-based research company in the field of information technology, the demand for systems designed to transform speech to text will only grow in the next 10 years. In recent years, the use of the End2End approach to automatic speech recognition has become increasingly popular. The main reason for this is the increasing size of computer databases and speech databases needed to train deep neural networks. In addition, models based on this approach and trained provide high results. This approach has several advantages, such as the phonetic alphabet required in the classical approach, a dictionary with word transcription, a large database of matching phonemes of the speech signal needed to teach the acoustic model, and even the absence of MFCC characters. At present, the high results obtained on the basis of transformers in the field of natural language processing (NLP) lead to the widespread use of this type of architectural networks in solving the problem of speech recognition.

II. THE MAIN PART

Although many approaches have been developed to address the problem of speech recognition at the moment, they do not provide the expected results. Therefore, new effective methods are emerging. One such effective method is the hybrid and integrated neural network-based (End-to-End) methods.

In hybrid approaches, the recognition system usually consists of several components, namely the acoustic model, the pronunciation model, and the language model. In this case, all components are trained independently and decoding graphs are constructed that perform the determination of the best transcription to obtain the final result. In the End-to-End approach, all parts of the speech recognition system are taught together. Therefore, such systems often output the resulting text faster than other approaches. End-to-End approaches are classified according to teaching criteria and structure. In recent years, solutions based on transformer neural networks have been providing better results than not only hybrid approaches, but also End-to-End approaches, as well as other modern solutions.

Transformers. Transformer architecture was first used in 2017 [1] to solve the problem of machine translation. This architecture and the principles of its operation are described in detail in the papers [1, 2] (Fig. 1).

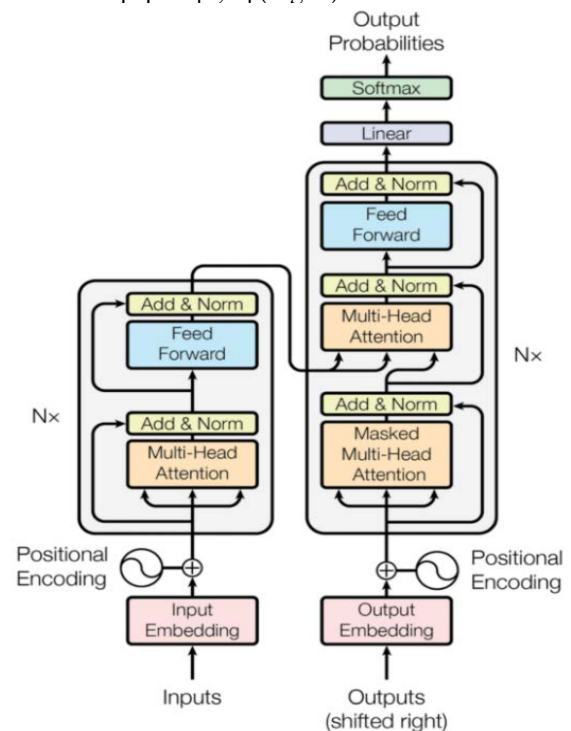


Fig. 1. Transformer architecture

Later, as a result of the evolution of the transformer architecture, natural language processing has made great strides in the field of NLP (Natural language processing). The range of problems that can be solved using neural networks in this architecture has expanded, and the results obtained have only improved. After high results in the field of NLP, transformers were applied to other areas of machine learning. Including speech recognition, speech synthesis, computer vision, and more.

Speech-Transformer. The use of transformers in speech recognition was carried out in 2018 by a group of Chinese scientists [2], and they proposed the use of the architecture shown in Figure 2.

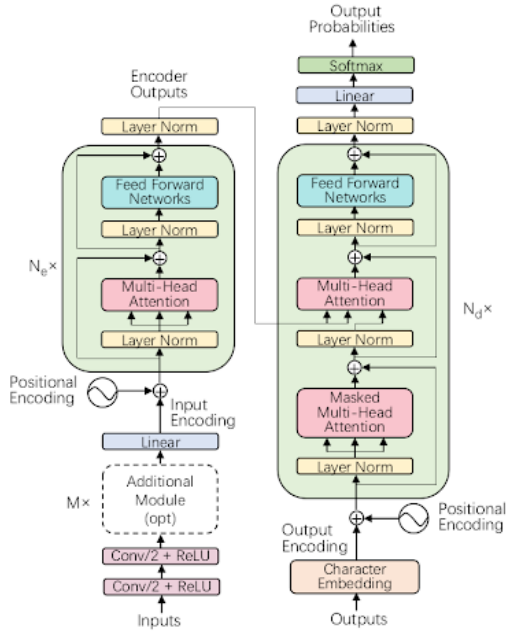


Fig. 2. Speech-Transformer architecture model

At the same time, minimal changes were made to the existing architecture, and a layer of wrapped neural networks (CNN-convolutional neural networks) was added to it before the input signals were transmitted to the transformer. This allows you to reduce the difference in input and output sizes (the number of frames in the audio varies greatly with the number of tokens in the text) and has a positive effect on the efficiency of the learning process.

Although the results obtained are not sufficient, this work shows that transformers can be used effectively in speech recognition. Since then, in 2019, the Speech-Transformer structure has undergone beneficial changes in several areas:

The integration of Speech-Transformer and CTC error functions has been proposed by the study authors [3]. The CTC error function is used in speech recognition, and it has a number of advantages. First, it allows you to consider the compatibility of certain audio frames to a specific character. In this case, the allowed alignments are made using the space symbol (blank). Second, this change serves to simplify the integration of the language model into the teaching process. As a result, sinusoidal positional encoding (PE) was abandoned. This is because the problems associated with long sequences are relevant to the speech recognition process. In [4] the work shifted from sinusoidal positional coding to relative coding, in [5] work was replaced by pool layers, and in [6] this coding phase was replaced by trained winding layers. In addition, in many papers published in

recent years, the advantages of such methods over the sinusoidal positional coding method have been proven. Sinusoidal position coding is performed using the following formula.

$$PE_{(pos, 2i)} = \sin\left(\frac{pos}{10000^{\frac{2i}{d}}}\right)$$

where pos - is the sequence number of the known token (word), d - is the size of the feature vector given in the algorithm for the token.

The first adaptations for current recognition of transformers were in the works [5] and [7], in which the encoder was initially adapted so that it could receive input data in block form and store it in a global context. The Monotonic Chunkwise Attention (MoChA) method was then used for online decoding.

For some systems (e.g., in hybrid approaches or transducer-based solutions), the acoustic model requires the same performance as the encoder. This allows the use of transformers in hybrid systems [8] as well as in transformer-based recognition systems [9].

A study published in 2019 [10] provided a comparative analysis of transformers with other approaches in the ESPNet (end-to-end speech processing toolkit) structure. At the same time, the recognition quality of transformer-based models has been confirmed. In 15 of the 13 cases, the models in the transformer architecture showed better results than the recurrent systems.

In late 2019 and early 2020, hybrid approaches to speech recognition approached the level of human recognition (SOTA-State-of-the-Art) [8]. As noted above, one of the key components of a system based on a hybrid approach to speech recognition is the acoustic model, and neural networks are currently used in its construction. [8] used several layers consisting of the encoder block of the transformer in the construction of the acoustic model. A schematic of one of these layers is shown in the figure below.

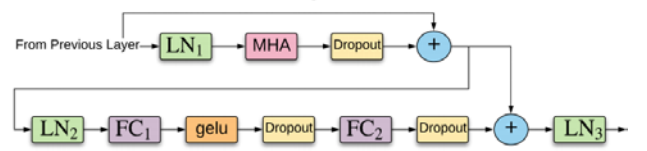


Fig. 3. Single transformer layer architecture

The mathematical expressions of the functions used in the blocks shown in Figure 3 are as follows:

LN_1, LN_2, LN_3 - is a layer normalization operation, and the existing shortcomings in the Batch Normalization have been eliminated in the work [22]. It is expressed as follows:

$$\mu^l = \frac{1}{H} \sum_{i=1}^H a_i^l \quad \sigma^l = \sqrt{\frac{1}{H} \sum_{i=1}^H (a_i^l - \mu^l)^2}$$

where H - is the number of neurons in the latent layer, μ^l - is the mean for the l^{th} layer, σ^l - is the variance value for the l^{th} layer, a_i^l - is the value of the i^{th} neuron in the l^{th} layer.

The *gelu* activation function is obtained as follows:

$$GELU(x) = xP(X \leq x) = x\Phi(x) = x \cdot \frac{1}{2} \left[1 + \operatorname{erf}\left(x / \sqrt{2}\right) \right]$$

If $X \sim N(0,1)$, that is, if the set values assigned to the input of this activating function are normalized between 0 and 1, then the above formula can be written as [23]:

$$0.5x \left(1 + \tanh \left[\sqrt{2/\pi} (x + 0.044715x^3) \right] \right)$$

FC_1, FC_2 - refers to a fully connected layer, *Dropout* - is the procedure of randomly dropping or disabling a portion of neurons in the hidden layers during training.

One of the most interesting aspects of this work is that the authors once again demonstrated that signal reduction through winding neural networks (neural networks such as VGG) is preferable to sinusoidal positional encoding, and used iterative loss to improve approximation in deep transformer network training.

Transformer Transducer. In late 2019 and the first half of 2020, researchers from Facebook [9] and Google [11] presented two models of neural networks with Transformer Transducer architecture. These models were created as a result of the integration of the transformer into the RNN-Transducer architecture.

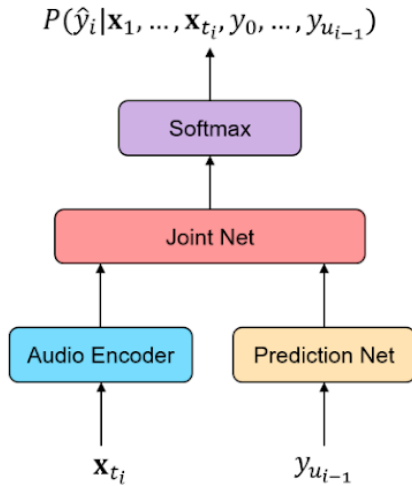


Fig. 4. Transducer architecture

The encoder part of the transformer was used for integration and is used instead of the audio encoder in the RNN-T structure. In [11], however, the predictive part of the model was also transformer-based and consisted of a small number of layers, as there was no need for a complex architecture. The difference between the RNN-T error function and the CTC is that the probabilities are based on the input sequence as well as the probabilities based on the prediction results. Also, since only the transformer encoder part is used in the Transformer Transducer architecture, it can be easily adapted for current recognition. [12] proposed an architecture in which the audio encoder part, called the Conv-Transformer Transducer, consisted of three blocks, in which each block consisted of a winding layer and a transformer layer. [13] proposed a variable context layer method by the authors in the study. This makes it possible to teach the model using a variable dimensional context.

Local and global context

One of the advantages of transformer-based architectures is that they take this global context into account with high efficiency. The importance of local connections in audio signals is greater than that of global connections, and in recent years several scientific papers have been published on the subject, detailing the advantages of transformers. [14] In the study, the authors proposed to add a wrapping layer after the Multi-Head Attention (MHA) block. Because this type of layer is effective in considering local dependencies. The resulting model was called Conformer by the authors. In this case, as in the Macaron-Net neural network, a half-step direct connection network was used.

[15] suggested a method of suppressing weak attention in the article. This ensures that weight values smaller than a given threshold value are set to zero, and that the model focuses on the most important frames in the context from a scattered focus throughout the context.

Unsupervised learning for describing a speech signal. Another area where transformers have found their application is the formation of high-level descriptions of the speech signal. In this case, the model is trained to recover spectrograms of unmarked speech signals after various distortions. Experiments have shown that even simple models give good results. Examples include Mockinjay [16], Speech-XLNet [17], Audio ALBERT [18], TERA [19], and wav2vec 2.0 [20]. Among these models, the wav2vec 2.0 model is an improved version of the vq-wav2vec [21] model, which transmits a set of signals from the winding layers to the transformer layer, and the contrast error function is used to teach the model.

In [22] authors of the work obtained high results from the ideas in wav2vec using the Conformer architecture. The use of LibriLight data for initial training of the model provided SOTA-level results in the LibriSpeech corpus.

Application in large-scale issues

In many scientific publications, the results of the models were mainly obtained in small bodies of around 1000 hours, such as LibriSpeech. However, in papers [23] and [24], transformer-based models have also been shown to have advantages over large-volume data.

III. EXPERIMENTS

Speech corpus. In order to teach the proposed neural network model, a speech corpus in Uzbek was created in the laboratory "Data Processing Systems". So far, several researches have been conducted in this laboratory on speech recognition and text analysis in Uzbek language[25-31]. The speeches consist of 432 hours of audiobooks and 72 hours of audio recordings of aphorisms and proverbs. The database was created by 174 speakers, and a series of audio files with a short duration (3-10 seconds) was initially used to teach the neural network model. A sequence of audio files lasting 6-9 min was provided for training after model approximation increased. Audio files have a sampling frequency of 16 kHz, and was stored in wav or mp3 formats contains a text file corresponding to them. The total size of the point database is 46 GB. 90% of the speech base created to teach the neural network model and the remaining 10% were allocated for testing.

Results on Uzbek speech corpus

Experimental research. An i7 processor, 32 GB of RAM, and a computer with a GTX 1660 Ti video card were purchased to train the Markov model and neural network models, as well as to conduct experimental research. Python programming language and Pytorch deep learning framework, Hidden Markov Model Toolkit (HTK) and ESPnet (end-to-end speech processing) tools were used in the construction and implementation of neural network models. The first model was based on the Markov model trained on 13 hours Uzbek speech dataset, which phonemes labeled by time, and then the general speech dataset was used. The second model consisting of 2 CNN layers and 2 BLSTM layers for the acoustic model, and 3 BLSTM layer network for the language model, were used. The third model consisted of 2 CNN layers and 3 BLSTM layers, using the CTC loss function in its training. In the proposed model, 2 half CNN layers and 1 transformer layer were used as the audio encoder. 2 transformer layers were used in the decoding part.

TABLE I. RESULTS OF EXPERIMENTS

Models	Traning time (hours)	Train accuracy (%)	Test accuracy (%)	Test error (WER) (%)
Hidden Markov Model based	63	85	69	31
Hybrid approach-based model	74	89	73	27
CNN+BLSTM+CTC loss	76	91	75	25
Transformer based Model	62	93	81	19

Researches have shown that the Transformer based Model is more efficient than other models, and this model was taken as the basis for the recognition system to be created [32].

CONCLUSION

An analysis of existing approaches has shown that transformer-based models in speech recognition are superior to other widely used models in terms of training time and accuracy. These models can be used to improve the quality and efficiency of speech recognition systems, as well as to add new natural languages to existing systems and to create recognition systems for them.

REFERENCES

- [1] A.Vaswani et al., 2017, "Attention Is All You Need", <https://arxiv.org/abs/1706.03762>
- [2] L.Dong et al., 2018 "Speech-Transformer: A No-Recurrence Sequence-to-Sequence Model for Speech Recognition", <https://ieeexplore.ieee.org/document/8462506>
- [3] S.Karita et al., 2019, "Improving Transformer-based End-to-End Speech Recognition with Connectionist Temporal Classification and Language Model Integration", <https://pdfs.semanticscholar.org/ffe1/416bcfde82f567dd280975bebcfeb4892298.pdf>
- [4] P.Zhou et al., 2019, "Improving Generalization of Transformer for Speech Recognition with Parallel Schedule Sampling and Relative Positional Embedding", <https://arxiv.org/abs/1911.00203>
- [5] E.Tsunoo et al., 2019, "Transformer ASR with Contextual Block Processing", <https://arxiv.org/abs/1910.07204>
- [6] A. Mohamed et al., 2019, "Transformers with convolutional context for ASR", <https://arxiv.org/abs/1904.11660>
- [7] E.Tsunoo et al., 2019, "Towards Online End-to-end Transformer Automatic Speech Recognition", <https://arxiv.org/abs/1910.11871>

- [8] Y.Wang et al., 2019, "Transformer-based Acoustic Modeling for Hybrid Speech Recognition", <https://arxiv.org/abs/1910.09799>
- [9] C.Yeh et al., 2019, "Transformer-Transducer: End-to-End Speech Recognition with Self-Attention", <https://arxiv.org/abs/1910.12977>
- [10] S.Karita et al., 2019, "A Comparative Study on Transformer vs RNN in Speech Applications", <https://arxiv.org/abs/1909.06317>
- [11] Q.Zhang et al., 2020, "Transformer Transducer: A Streamable Speech Recognition Model with Transformer Encoders and RNN-T Loss", <https://arxiv.org/abs/2002.02562>
- [12] W.Huang et al., 2020, "Conv-Transformer Transducer: Low Latency, Low Frame Rate, Streamable End-to-End Speech Recognition", <https://arxiv.org/abs/2008.05750>
- [13] A.Tripathi et al., 2020, "Transformer Transducer: One Model Unifying Streaming and Non-streaming Speech Recognition", <https://arxiv.org/abs/2010.03192>
- [14] A.Gulati et al., 2020, "Conformer: Convolution-augmented Transformer for Speech Recognition", <https://arxiv.org/abs/2005.08100>
- [15] Y.Shi et al., 2020, "Weak-Attention Suppression For Transformer Based Speech Recognition", <https://arxiv.org/abs/2005.09137>
- [16] A.T.Liu et al., 2019, "Mockingjay: Unsupervised Speech Representation Learning with Deep Bidirectional Transformer Encoders", <https://arxiv.org/abs/1910.12638>
- [17] X.Song et al., 2020, "Speech-XLNet: Unsupervised Acoustic Model Pretraining For Self-Attention Networks", <https://arxiv.org/abs/1910.10387>
- [18] P.Chi et al., 2020, "Audio ALBERT: A Lite BERT for Self-supervised Learning of Audio Representation", <https://arxiv.org/abs/2005.08575>
- [19] A.T.Liu et al., 2020, "TERA: Self-Supervised Learning of Transformer Encoder Representation for Speech", <https://arxiv.org/abs/2007.06028>
- [20] A.Baevski et al., 2020, "wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations", <https://arxiv.org/abs/2006.11477>
- [21] A.Baevski et al., 2020 "vq-wav2vec: Self-Supervised Learning of Discrete Speech Representations", <https://arxiv.org/abs/1910.05453>
- [22] Y.Zhang et al., 2020, "Pushing the Limits of Semi-Supervised Learning for Automatic Speech Recognition", <https://arxiv.org/abs/2010.10504>
- [23] L.Lu et al., 2020, "Exploring Transformers for Large-Scale Speech Recognition", <https://arxiv.org/abs/2005.09684>
- [24] Y.Wang et al., 2020, "Transformer in action: a comparative study of transformer-based acoustic models for large scale speech recognition applications", <https://arxiv.org/abs/2010.14665>
- [25] Babomuradov Ozod, O. J., Mamatov, N. S., Boboev, L. B., & Otaxonova, B. (2019). Text documents classification in Uzbek language. International Journal of Recent Technology and Engineering, 8(2 Special Issue 11), 3787–3789. <https://doi.org/10.35940/ijrte.B1493.0982S1119>
- [26] Mamatov, N. S., Samijonov, A. N., Yuldoshev, Y., & Khusan, R. (2020). Selection the informative features on the basis of interrelationship of features. In Techno-Societal 2018 - Proceedings of the 2nd International Conference on Advanced Technologies for Societal Applications (Vol. 2, pp. 121–129). https://doi.org/10.1007/978-3-030-16962-6_13
- [27] Fazilov, S., Mamatov, N., Samijonov, A., & Abdullaev, S. (2020). Reducing the dimensionality of feature space in pattern recognition tasks. Journal of Physics: Conference Series, 1441(1), 012139. <https://doi.org/10.1088/1742-6596/1441/1/012139>
- [28] Mamatov, N., Samijonov, A., & Niyozmatova, N. (2020). Determination of non-informative features based on the analysis of their relationships. Journal of Physics: Conference Series, 1441(1), 012149. <https://doi.org/10.1088/1742-6596/1441/1/012149>
- [29] Niyozmatova, N. A., Mamatov, N., Samijonov, A., Mamadalieva, N., & Abdullayeva, B. M. (2020). Unconditional discrete optimization of linear-fractional function "–1"-order. IOP Conference Series: Materials Science and Engineering, 862(4), 042028. <https://doi.org/10.1088/1757-899X/862/4/042028>
- [30] Samijonov, A., Mamatov, N., Niyozmatova, N. A., Yuldoshev, Y., & Asraev, M. (2020). Gradient method for determining non-informative features on the basis of a homogeneous criterion with a positive

- degree. IOP Conference Series: Materials Science and Engineering, 919(4). <https://doi.org/10.1088/1757-899X/919/4/042011>
- [31] Niyozmatova, N. A., Mamatov, N., Samijonov, A., Rahmonov, E., & Juraev, S. (2020). Method for selecting informative and non-informative features. IOP Conference Series: Materials Science and Engineering, 919(4). <https://doi.org/10.1088/1757-899X/919/4/042013>
- [32] Mamatov, N., Niyozmatova, N. A., Samijonov, A., Juraev, S., & Abdullayeva, B. (2020). The choice of informative features based on heterogeneous functionals. IOP Conference Series: Materials Science and Engineering, 919(4). <https://doi.org/10.1088/1757-899X/919/4/042009>
- [33] Fazilov, S., & Mamatov, N. (2019). Formation an informative description of recognizable objects. Journal of Physics: Conference Series, 1210(1). <https://doi.org/10.1088/1742-6596/1210/1/012043>
- [34] Mamatov, N., Samijonov, A., & Yuldashev, Z. (2019). Selection of features based on relationships. Journal of Physics: Conference Series, 1260(10), 102008. <https://doi.org/10.1088/1742-6596/1260/10/102008>
- [35] Shavkat, F., Narzillo, M., & Abdurashid, S. (2019). Selection of significant features of objects in the classification data processing. International Journal of Recent Technology and Engineering, 8(2 Special Issue 11), 3790–3794. <https://doi.org/10.35940/ijrte.B1494.0982S1119>
- [36] Mamatov, N., Samijonov, A., Yuldashev, Z., & Niyozmatova, N. (2019). Discrete Optimization of Linear Fractional Functionals. 2019 15th International Asian School-Seminar Optimization Problems of Complex Systems, OPCS 2019, 96–99. <https://doi.org/10.1109/OPCS.2019.8880208>
- [37] Shavkat, F., Narzillo, M., & Nilufar, N. (2019). Developing methods and algorithms for forming of informative features' space on the base K-types uniform criteria. International Journal of Recent Technology and Engineering, 8(2 Special Issue 11), 3784–3786. <https://doi.org/10.35940/ijrte.B1492.0982S1119>